

GrounUp Enterprise Specification

Phase 18 - Enterprise Scheduling & Resource Planning

Version 1.0.0 / Review-ready / Proposed lock: GES-P18-v1.0.0

1. Mission

Create a governed planning and scheduling platform that converts scope, quantities, resources, commitments and field progress into executable project and portfolio schedules with transparent logic, capacity and forecast consequences.

2. Architecture Principles

Reproducible CPM logic; immutable approved baselines; explicit calendars/constraints; resource-capacity awareness; production-based progress; integrated cost/schedule; scenario transparency; field planning integration; human-governed AI.

3. Cross-Phase Integration

Phase 18 consumes estimating quantities/production assumptions, project operations, workforce availability, finance/cost, procurement/material lead times, fleet availability and Phase 13 analytics. It also exchanges field progress, resource demand, milestones and forecasts with those phases.

4. CPM & Baselines

Activities, relationships, lags, constraints and calendars drive reproducible forward/backward calculations, float, critical path and longest path. Approved baselines remain immutable; rebaselines create new controlled versions and preserve prior states.

5. Resource Planning

Labor, crews, equipment, trucks, materials and subcontractors are loaded against time-phased activities. Capacity is constrained by actual availability and commitments rather than assumed unlimited.

6. Look-Ahead, Last Planner & Pull Planning

Near-term planning includes readiness checks, constraint removal, weekly commitments, PPC, reasons for variance, collaborative pull planning and handoff buffers.

7. Progress & Forecasting

Updates capture status date, actual starts/finishes, remaining duration, quantities, physical progress and production. Forecast completion and milestones are recalculated and compared against baseline/current commitments.

8. Risk, Scenario & Recovery

The platform supports duration uncertainty, risk events, confidence dates, sensitivities, alternative crew/equipment/calendar scenarios, recovery plans, crashing and fast-tracking with cost/resource impacts.

9. Delay & Impact Records

Delay events, impacted activities, fragnets and time-impact-analysis records preserve contemporaneous schedule evidence and assumptions. They support later contract/claims workflows but do not generate legal conclusions by themselves.

10. AI Schedule Intelligence

AI may detect likely constraints, predict delay risk, draft narratives and recommend recovery options. Outputs must identify supporting schedule/resource evidence and confidence and require human approval for consequential changes.

11. Functional Domains

Enterprise Scheduling Core

Enterprise, portfolio, project and work-package schedules with baselines, calendars, revisions and audit.

CPM Scheduling

Activities, relationships, constraints, float, critical path, longest path, lags and schedule calculation.

Work Breakdown Structure

Program, project, phase, area, discipline, cost-code and work-package decomposition.

Calendars & Shifts

Company, project, crew, equipment, holiday, weather and shift calendars.

Resource Planning

Labor, crews, equipment, trucks, subcontractors, materials and capacity requirements.

Resource Loading

Time-phased resource demand, histograms, utilization, overloads and allocation.

Resource Leveling

Constraint-aware leveling, smoothing, priority rules, alternative scenarios and approvals.

Crew Planning

Crew templates, crew-size recommendations, craft mix, shift plans, availability and handoff to operations.

Equipment Planning

Equipment class/model demand, availability, rentals, utilization and fleet integration.

Truck & Haul Planning

Truck demand, cycle assumptions, haul windows, dispatch capacity and schedule coupling.

Material Planning

Required-on-site dates, lead times, procurement milestones, delivery windows and inventory availability.

Subcontractor Planning

Trade capacity, commitments, mobilization, sequencing, access windows and coordination.

Look-Ahead Planning

2/3/6-week look-aheads, constraints, commitments, readiness and weekly planning.

Last Planner & Commitments

Weekly work planning, promises, percent plan complete, reasons for variance and learning.

Pull Planning

Milestone-driven pull sessions, handoffs, buffers, promises, constraints and collaborative sequencing.

Milestones & Contract Dates

Notice-to-proceed, interim milestones, substantial/final completion and contractual date control.

Baseline Management

Original/current baselines, approved revisions, comparison, change reasons and governance.

Schedule Update Cycle

Status date, actual starts/finishes, remaining duration, physical progress and update approvals.

Progress Measurement

Quantity, duration, units-complete, weighted steps, earned value and production-based progress.

Cost & Schedule Integration

Budget, cost codes, earned value, cash flow, commitments and forecast integration.

Constraint Management

Design, permit, RFI, submittal, procurement, access, utility, environmental and resource constraints.

Weather & Calendar Impacts

Observed/forecast weather references, non-work days, productivity impacts and schedule scenarios.

Schedule Risk Analysis

Uncertainty ranges, risk events, sensitivity, confidence dates and schedule contingency.

Recovery & Acceleration

Recovery plans, resequencing, added shifts/crews/equipment, crashing, fast-tracking and cost impacts.

Delay & Impact Analysis

Change/delay events, impacted activities, fragnet references, contemporaneous records and claims support.

Portfolio Capacity Planning

Cross-project labor/equipment capacity, backlog demand, bottlenecks and strategic resource forecasting.

What-If & Scenario Planning

Alternative sequences, crew /equipment levels, calendars, procurement dates and completion outcomes.

AI Schedule Intelligence

Evidence-grounded delay prediction, constraint detection, recovery recommendations and narrative.

Scheduling Digital Twin

Current schedule, resources, constraints, production, costs, risks and decision state.

12. Controlled Artifact Summary

Artifact	Count
Requirements	400
Entities	185
Fields	300
Business Rules	80
Validation Rules	80
Workflows	52
Engines	59
Api Resources	64
Ui Surfaces	49
Reports	42
Kpis	52
Tests	400

13. KPI Registry

P18-KPI-001 - Schedule Performance Index: Governed metric definition for Schedule Performance Index.

P18-KPI-002 - Milestone On-Time %: Governed metric definition for Milestone On-Time %.

P18-KPI-003 - Critical Activity On-Time %: Governed metric definition for Critical Activity On-Time %.

P18-KPI-004 - Critical Path Duration: Governed metric definition for Critical Path Duration.

P18-KPI-005 - Total Float Trend: Governed metric definition for Total Float Trend.

P18-KPI-006 - Negative Float Count: Governed metric definition for Negative Float Count.

P18-KPI-007 - Open Constraints: Governed metric definition for Open Constraints.

P18-KPI-008 - Constraint Closure On-Time %: Governed metric definition for Constraint Closure On-Time %.

P18-KPI-009 - Look-Ahead Readiness %: Governed metric definition for Look-Ahead Readiness %.

P18-KPI-010 - PPC %: Governed metric definition for PPC %.

P18-KPI-011 - Reason-for-Variance Closure: Governed metric definition for Reason-for-Variance Closure.

P18-KPI-012 - Baseline Change Count: Governed metric definition for Baseline Change Count.

P18-KPI-013 - Schedule Update Timeliness: Governed metric definition for Schedule Update Timeliness.

P18-KPI-014 - Actual Progress vs Planned: Governed metric definition for Actual Progress vs Planned.

P18-KPI-015 - Quantity Progress vs Planned: Governed metric definition for Quantity Progress vs Planned.

P18-KPI-016 - Production Rate vs Plan: Governed metric definition for Production Rate vs Plan.

P18-KPI-017 - Forecast Completion Variance: Governed metric definition for Forecast Completion Variance.

P18-KPI-018 - Forecast Milestone Variance: Governed metric definition for Forecast Milestone Variance.

P18-KPI-019 - Labor Demand Variance: Governed metric definition for Labor Demand Variance.

P18-KPI-020 - Crew Utilization: Governed metric definition for Crew Utilization.

P18-KPI-021 - Crew Overload Hours: Governed metric definition for Crew Overload Hours.

P18-KPI-022 - Equipment Demand Variance: Governed metric definition for Equipment Demand Variance.

P18-KPI-023 - Equipment Utilization: Governed metric definition for Equipment Utilization.

P18-KPI-024 - Equipment Overload Hours: Governed metric definition for Equipment Overload Hours.

P18-KPI-025 - Truck Demand Variance: Governed metric definition for Truck Demand Variance.

P18-KPI-026 - Material Need-Date Compliance: Governed metric definition for Material Need-Date Compliance.

P18-KPI-027 - Subcontractor Capacity Coverage: Governed metric definition for Subcontractor Capacity Coverage.

P18-KPI-028 - Resource Leveling Exception Count: Governed metric definition for Resource Leveling Exception Count.

P18-KPI-029 - Resource Smoothing Improvement: Governed metric definition for Resource Smoothing Improvement.

P18-KPI-030 - Weather Impact Days: Governed metric definition for Weather Impact Days.

P18-KPI-031 - Schedule Risk P80 Finish: Governed metric definition for Schedule Risk P80 Finish.

P18-KPI-032 - Schedule Risk Exposure: Governed metric definition for Schedule Risk Exposure.

P18-KPI-033 - Recovery Plan Closure: Governed metric definition for Recovery Plan Closure.

P18-KPI-034 - Acceleration Cost per Day Saved: Governed metric definition for Acceleration Cost per Day Saved.

P18-KPI-035 - Delay Event Aging: Governed metric definition for Delay Event Aging.

P18-KPI-036 - TIA Cycle Time: Governed metric definition for TIA Cycle Time.

P18-KPI-037 - Earned Value Variance: Governed metric definition for Earned Value Variance.

P18-KPI-038 - Cost Performance Index: Governed metric definition for Cost Performance Index.

P18-KPI-039 - Cash Flow Forecast Variance: Governed metric definition for Cash Flow Forecast Variance.

P18-KPI-040 - Portfolio Capacity Gap: Governed metric definition for Portfolio Capacity Gap.

P18-KPI-041 - Backlog Coverage: Governed metric definition for Backlog Coverage.

P18-KPI-042 - Critical Resource Bottleneck Count: Governed metric definition for Critical Resource Bottleneck Count.

P18-KPI-043 - Schedule Quality Score: Governed metric definition for Schedule Quality Score.

P18-KPI-044 - Open Logic Exceptions: Governed metric definition for Open Logic Exceptions.

P18-KPI-045 - Open Constraint Exceptions: Governed metric definition for Open Constraint Exceptions.

P18-KPI-046 - AI Delay Prediction Precision: Governed metric definition for AI Delay Prediction Precision.

P18-KPI-047 - AI Constraint Finding Acceptance: Governed metric definition for AI Constraint Finding Acceptance.

P18-KPI-048 - AI Recovery Recommendation Acceptance: Governed metric definition for AI Recovery Recommendation Acceptance.

P18-KPI-049 - AI Evidence Coverage: Governed metric definition for AI Evidence Coverage.

P18-KPI-050 - Schedule Dashboard Adoption: Governed metric definition for Schedule Dashboard Adoption.

P18-KPI-051 - Scheduling Digital Twin Freshness: Governed metric definition for Scheduling Digital Twin Freshness.

P18-KPI-052 - Schedule Decision Cycle Time: Governed metric definition for Schedule Decision Cycle Time.

14. Engine Registry

P18-ENG-001 - CPM Calculation Engine (Deterministic): Provides cpm calculation engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-002 - Forward Pass Engine (Deterministic): Provides forward pass engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-003 - Backward Pass Engine (Deterministic): Provides backward pass engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-004 - Float Calculation Engine (Deterministic): Provides float calculation engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-005 - Critical Path Engine (Deterministic): Provides critical path engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-006 - Longest Path Engine (Deterministic): Provides longest path engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-007 - Relationship Validation Engine (Deterministic): Provides relationship validation engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-008 - Constraint Validation Engine (Deterministic): Provides constraint validation engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-009 - Calendar Engine (Deterministic): Provides calendar engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-010 - Shift Calendar Engine (Deterministic): Provides shift calendar engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-011 - Baseline Comparison Engine (Deterministic): Provides baseline comparison engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-012 - Schedule Update Engine (Deterministic): Provides schedule update engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-013 - Progress Measurement Engine (Deterministic): Provides progress measurement engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-014 - Quantity Progress Engine (Deterministic): Provides quantity progress engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-015 - Production Progress Engine (Deterministic): Provides production progress engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-016 - Earned Value Engine (Deterministic): Provides earned value engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-017 - Crew Demand Engine (Deterministic): Provides crew demand engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-018 - Equipment Demand Engine (Deterministic): Provides equipment demand engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-019 - Truck Demand Engine (Deterministic): Provides truck demand engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-020 - Material Need-Date Engine (Deterministic): Provides material need-date engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-021 - Subcontractor Demand Engine (Deterministic): Provides subcontractor demand engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-022 - Resource Loading Engine (Deterministic): Provides resource loading engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-023 - Resource Histogram Engine (Deterministic): Provides resource histogram engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-024 - Resource Leveling Engine (Deterministic): Provides resource leveling engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-025 - Resource Smoothing Engine (Deterministic): Provides resource smoothing engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-026 - Capacity Gap Engine (Deterministic): Provides capacity gap engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-027 - Look-Ahead Engine (Deterministic): Provides look-ahead engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-028 - Weekly Work Plan Engine (Deterministic): Provides weekly work plan engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-029 - PPC Engine (Deterministic): Provides ppc engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-030 - Constraint Readiness Engine (Deterministic): Provides constraint readiness engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-031 - Pull Planning Engine (Deterministic): Provides pull planning engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-032 - Weather Impact Engine (Deterministic): Provides weather impact engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-033 - Forecast Completion Engine (Deterministic): Provides forecast completion engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-034 - Milestone Forecast Engine (Deterministic): Provides milestone forecast engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-035 - Scenario Engine (Deterministic): Provides scenario engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-036 - What-If Engine (Deterministic): Provides what-if engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-037 - Schedule Risk Engine (Deterministic): Provides schedule risk engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-038 - Confidence Date Engine (Deterministic): Provides confidence date engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-039 - Sensitivity Engine (Deterministic): Provides sensitivity engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-040 - Recovery Planning Engine (Deterministic): Provides recovery planning engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-041 - Acceleration Engine (Deterministic): Provides acceleration engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-042 - Crash Analysis Engine (Deterministic): Provides crash analysis engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-043 - Fast-Track Analysis Engine (Deterministic): Provides fast-track analysis engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-044 - Delay Impact Engine (Deterministic): Provides delay impact engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-045 - Fragnet Engine (Deterministic): Provides fragnet engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-046 - Time Impact Analysis Engine (Deterministic): Provides time impact analysis engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-047 - Portfolio Capacity Engine (Deterministic): Provides portfolio capacity engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-048 - Backlog Demand Engine (Deterministic): Provides backlog demand engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-049 - Cash Flow Curve Engine (Deterministic): Provides cash flow curve engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-050 - Schedule-Cost Integration Engine (Deterministic): Provides schedule-cost integration engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-051 - Schedule Quality Engine (Deterministic): Provides schedule quality engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-052 - Schedule Alert Engine (Deterministic): Provides schedule alert engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-053 - Schedule KPI Engine (Deterministic): Provides schedule kpi engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-054 - Delay Prediction Engine (AI-assisted): Provides delay prediction engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-055 - AI Constraint Detection Engine (AI-assisted): Provides ai constraint detection engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-056 - AI Recovery Recommendation Engine (AI-assisted): Provides ai recovery recommendation engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-057 - Evidence Retrieval Engine (AI-assisted): Provides evidence retrieval engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-058 - AI Confidence Engine (AI-assisted): Provides ai confidence engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

P18-ENG-059 - Scheduling Digital Twin Engine (AI-assisted): Provides scheduling digital twin engine using governed schedule logic, calendars, resource data, progress and cross-phase evidence.

15. Workflow Registry

P18-WF-001 - Project Schedule Creation: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-002 - WBS Creation: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-003 - Calendar Setup: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-004 - Activity & Logic Development: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-005 - Milestone Setup: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-006 - Schedule Quality Check: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-007 - Initial Baseline Approval: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-008 - Baseline Revision: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-009 - Schedule Update Cycle: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-010 - Status Date Advance: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-011 - Actual Progress Update: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-012 - Quantity-Based Progress Update: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-013 - Production-Based Progress Update: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-014 - Resource Loading: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-015 - Labor/Crew Loading: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-016 - Equipment Loading: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-017 - Truck/Haul Loading: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-018 - Material Need-Date Planning: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-019 - Subcontractor Loading: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-020 - Resource Leveling: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-021 - Resource Smoothing: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-022 - Look-Ahead Creation: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-023 - Weekly Work Planning: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-024 - Commitment/PPC Review: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-025 - Pull Planning Session: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-026 - Constraint Identification: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-027 - Constraint Escalation: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-028 - Constraint Closure: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-029 - Weather Impact Update: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-030 - Schedule Risk Analysis: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-031 - Scenario Creation: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-032 - What-If Comparison: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-033 - Recovery Plan: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-034 - Acceleration Review: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-035 - Design Change Impact: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-036 - Delay Event Capture: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-037 - Fragnet Creation: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-038 - Time Impact Analysis: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-039 - Portfolio Capacity Planning: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-040 - Backlog Resource Forecast: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-041 - Cash Flow/Schedule Integration: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-042 - Earned Value Update: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-043 - Forecast Completion Update: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-044 - Critical Path Review: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-045 - Longest Path Review: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-046 - Schedule Narrative: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-047 - Executive Schedule Review: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-048 - Schedule Alert & Escalation: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-049 - External Schedule Import: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-050 - Schedule Export: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-051 - Scheduling Digital Twin Refresh: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

P18-WF-052 - Schedule Closeout & Archive: Initiate > Validate > Calculate/Plan > Review > Approve > Publish/Execute > Monitor > Close

16. Acceptance & Lock Gate

Each controlled requirement maps to a test case and implementation surface. Phase 18 remains review-ready until Project Controls/Scheduling, Operations, Finance, Workforce, Fleet, Procurement, Architecture, Security/Privacy, AI Governance, Data Governance, integration and UAT approvals are complete.